

Analyzing Performance

Understanding a company's past is essential to forecasting its future, so a thorough analysis of historical performance is a critical component of valuation. Always begin with the core elements of value creation: return on invested capital (ROIC) and revenue growth. Examine trends in the company's long-run performance and its performance relative to that of its peers, so you can base your forecasts of future cash flows on reasonable assumptions about the company's key value drivers.

Start by analyzing ROIC, both with and without goodwill. ROIC with goodwill measures the company's ability to create value over and above premiums paid for acquisitions. ROIC without goodwill is a better measure of the company's underlying operating performance compared with that of its peers. Then drill down into the components of ROIC to build an integrated view of the company's operating performance and understand which aspects of the business are responsible for its overall performance. Next, examine what drives revenue growth. Does revenue growth stem, for instance, more from organic growth or from currency effects, which are largely beyond management control and probably not sustainable? Finally, assess the company's financial health to determine whether it has the financial resources to conduct business and make investments for growth.

ANALYZING RETURNS ON INVESTED CAPITAL

Chapter 11 reorganized the income statement into net operating profit after taxes (NOPAT) and the balance sheet into invested capital. ROIC measures the ratio of NOPAT to invested capital:

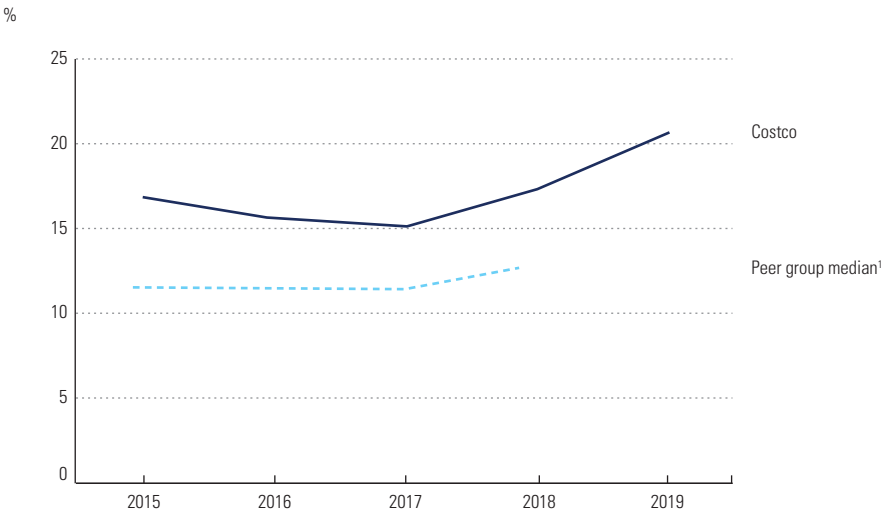
$$\text{ROIC} = \frac{\text{NOPAT}}{\text{Invested Capital}}$$

Companies that report ROIC in their annual reports may compute it using starting invested capital, ending capital, or the average of the two. Since profit is measured over an entire year, whereas capital is measured only at one point in time, we recommend that you average starting and ending invested capital. If the business is highly seasonal, such that capital is changing substantially at the company's fiscal close, consider using quarterly averages.

ROIC is a better analytical tool than return on equity (ROE) or return on assets (ROA) for understanding the company's performance because it focuses solely on a company's operations. ROE mixes operating performance with capital structure, making peer-group analysis and trend analysis less insightful. ROA—even when calculated on a pre-interest basis—is an inadequate measure of performance because it includes nonoperating assets and ignores the benefits of accounts payable and other operating liabilities that together reduce the amount of capital required from investors.

As an example of using ROIC to analyze performance, Exhibit 12.1 plots ROIC for Costco and the median of its peers from 2015 to 2019, based on the NOPAT and invested-capital calculations presented in Chapter 11.¹ Costco has consistently earned higher returns on invested capital than its peers, and

EXHIBIT 12.1 **Costco versus Peer Group: Return on Invested Capital**



¹ ROIC measured on average capital without goodwill and acquired intangibles.

² For peers, 2019 results were not available at the time of this writing. Costco's fiscal year ended September 1, 2019, versus December 2019 to January 2020 for peers.

¹ Costco's fiscal year ends on the Sunday closest to August 31, so its 2019 fiscal year ended September 1, 2019. Its peers end their fiscal years in December or January, and their 2019 results were not available at the time of this writing.

showed significant increases in 2018 and 2019. As we will show later, Costco's higher ROIC can be traced to its lower operating profit margin offset by strong capital productivity.

Analyzing ROIC with and without Goodwill and Acquired Intangibles

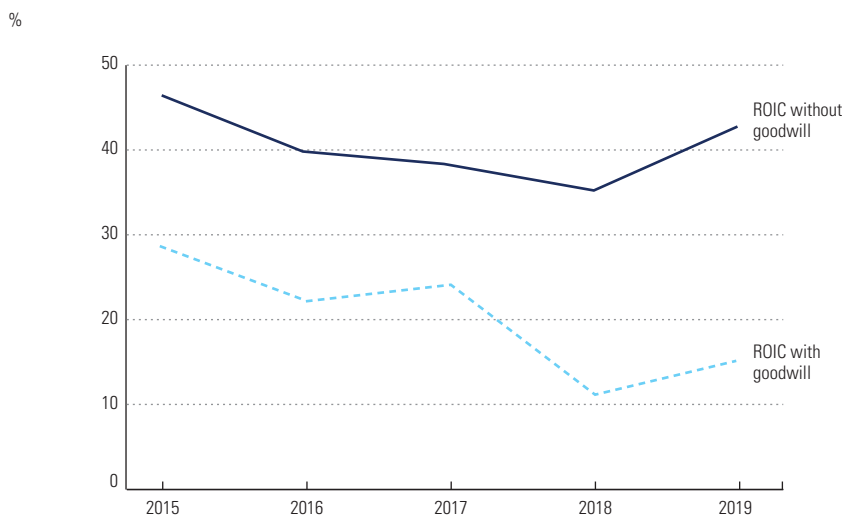
Goodwill and acquired intangibles are intangible assets purchased in an acquisition. ROIC should be computed both with and without goodwill and acquired intangibles. In our analysis, we treat goodwill identically to acquired intangibles.² Therefore, we will often shorten the expression *goodwill and acquired intangibles* to simply *goodwill*.

The reason to compute ROIC with and without goodwill is that each ratio analyzes different things. ROIC with goodwill measures whether the company has earned adequate returns for shareholders, factoring in the price paid for acquisitions. ROIC excluding goodwill measures the underlying operating performance of a company. It tells you whether the underlying economics generate ROIC above the cost of capital. It can be used to compare a company's performance against that of peers and to analyze trends. It is not affected by the price premiums paid for acquisitions. ROIC without goodwill is also more relevant for projecting future cash flows and setting strategy. A company does not need to spend more on acquisitions to grow organically, so ROIC without goodwill is a more relevant baseline for forecasting cash flows. Finally, companies that have a high ROIC without goodwill will likely create more value from growth, while companies that have low ROIC without goodwill will likely create more value by improving ROIC.

Costco doesn't have any goodwill, having grown entirely organically, but for companies that make significant acquisitions, the difference between ROIC with and without goodwill can be large. Exhibit 12.2 presents ROIC with and without goodwill for the U.S. luxury-goods maker Tapestry, formerly known as Coach. In 2017, the company purchased Kate Spade, a luxury fashion design house, for \$2.4 billion in cash. Since the two companies had quite similar returns on capital, Tapestry's ROIC without goodwill remained fairly constant before and after the acquisition. In contrast, ROIC with goodwill after the acquisition fell from 24 percent to 12 percent in 2017. Does the decline in ROIC when measured with goodwill imply that the acquisition destroyed value? Not necessarily: cost savings and cross-selling opportunities take time to realize, and Tapestry's access to new customers, especially millennials, may accelerate growth in its own product lines.

² To be classified as an acquired intangible, the asset must be separable and identifiable, as in the case of patents. Goodwill describes assets that are not separable or identifiable. Acquired intangibles are amortized over the life of the asset, whereas goodwill is impaired if value falls below book value. Since we analyze the two accounts in the same manner, we do not make a distinction.

EXHIBIT 12.2 Tapestry: Return on Invested Capital



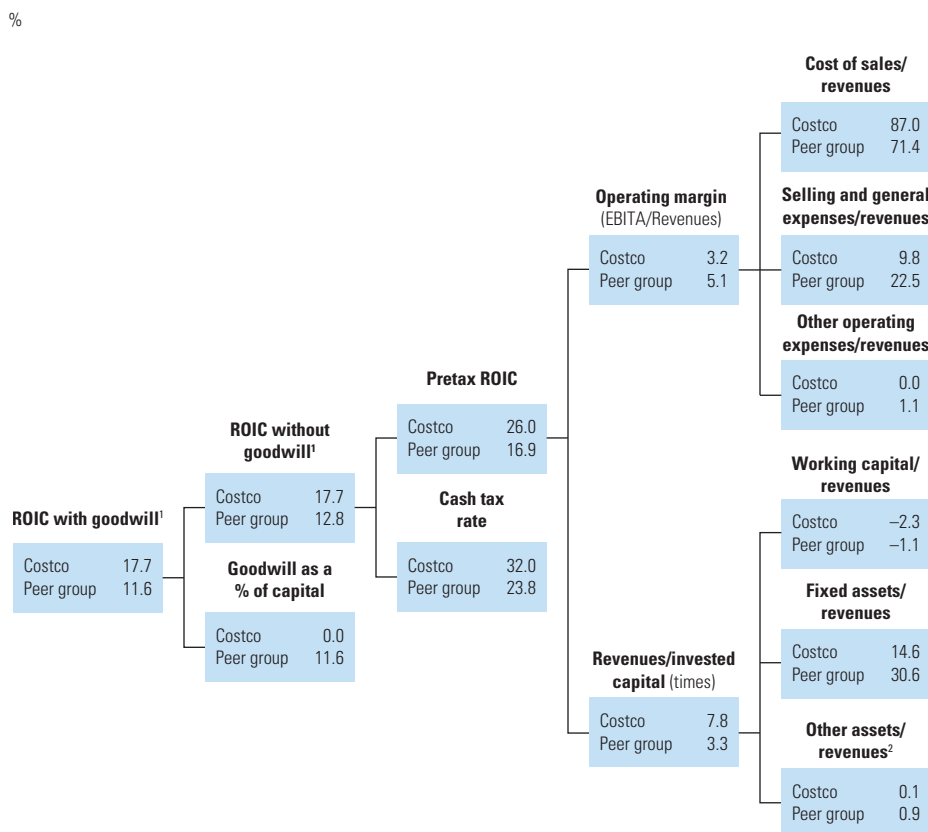
Accurately evaluating ROIC with goodwill leads to a second challenge: ROIC may increase even without improvements to the underlying business. We've seen situations where a business unit submitted a new strategic plan saying it expected to improve its ROIC over time. On the surface, its forecast looked impressive, but we then discovered that the ROIC included goodwill, and the expected improvement in ROIC would be caused solely by goodwill remaining constant as the business grew profits organically. The management team would earn accolades for improving ROIC purely as a result of the accounting for goodwill, not an underlying improvement to the business.

Decomposing ROIC to Develop an Integrated Perspective of Company Economics

To show how we analyze a company's economics based on decomposition of its ROIC, we return to the example of Costco and its peers. Costco has consistently earned a higher ROIC than its peers. But what caused this difference in performance? To understand which elements of a company's business are driving the company's ROIC, split apart the ratio as follows:

$$\text{ROIC} = (1 - \text{Operating Cash Tax Rate}) \times \frac{\text{EBITA}}{\text{Revenues}} \times \frac{\text{Revenues}}{\text{Invested Capital}}$$

The preceding equation is one of the most powerful equations in financial analysis. It demonstrates the extent to which a company's ROIC is driven by

EXHIBIT 12.3 **Costco versus Peer Group: ROIC Tree, 2018**

¹ To match economic-profit valuation, invested capital is measured at beginning of year.

² Other assets, net other liabilities.

its ability to maximize profitability (EBITA divided by revenues, or the operating margin), optimize capital turnover (measured by revenues over invested capital), or minimize operating taxes.

Each of these components can be further disaggregated, so that each expense and capital item can be analyzed, line item by line item. Exhibit 12.3 shows how the components can be organized into a tree. On the right side of the tree are operational financial ratios, the drivers of value over which managers have control. Reading from right to left, each subsequent box is a function of the boxes to its right. For example, operating margin equals 100 percent less the ratios of cost of sales to revenues, selling and general expenses to revenues, and other operating expenses to revenues. Pretax ROIC equals operating margin times capital turnover (revenues divided by invested capital), and so on.

Once you have calculated the historical drivers of ROIC, compare them with the ROIC drivers of other companies in the same industry. You can then weigh this perspective against your analysis of the industry structure (opportunities for differentiation, barriers to entry or exit, etc.) and a qualitative assessment of the company's strengths and weaknesses.

To illustrate, let's examine the difference between Costco and its peers. In 2018, Costco's ROIC with goodwill equaled 17.7 percent, compared with its peers' median of 11.6 percent. The difference is somewhat smaller without goodwill, because Costco had no goodwill. You might ask what drives Costco's higher ROIC. Costco has an unusual business model for a retailer. It doesn't mark up its costs as much as other retailers, leading to a higher cost of sales relative to revenues. It makes up for that with lower selling and general expenses. For example, its warehouse format has much lower depreciation, and its cost to stock shelves is lower because it doesn't put items on the shelves individually but instead uses the manufacturers' containers. Costco also sells larger sizes of its products with a smaller assortment to manage. Despite the lower selling and general expenses, it still ends up with a lower operating profit margin (3.2 percent, versus 5.1 percent). It makes up for this with higher capital productivity—primarily much lower fixed assets relative to sales.

Line Item Analysis A comprehensive valuation model will convert every line item in the company's financial statements into some type of ratio. For the income statement, most items are taken as a percentage of sales. (Exceptions exist: operating cash taxes, for instance, should be calculated as a percentage of pretax operating profits, not as a percentage of sales.)

For the balance sheet, each line item can also be taken as a percentage of revenues (or as a percentage of cost of goods sold for inventories and payables, to avoid distortion caused by changing prices). For operating current assets and liabilities, you can also convert each line item into days, using the following formula:

$$\text{Days} = 365 \times \frac{\text{Balance Sheet Item}}{\text{Revenues}}$$

If the business is seasonal, operating ratios such as inventories should be calculated using quarterly data. The differences can be quite substantial.

The use of days lends itself to a simple operational interpretation. How much cash is tied up in the business, and for how long? As Exhibit 12.4 demonstrates, Costco and its peers have negative working capital, with Costco's somewhat lower. Costco's product selection and business model results in lower levels of inventory and accounts payable. In 2018, it had only 30.9 days of inventory, versus 52.7 for its peers. In other words, goods don't stay on Costco's shelves

EXHIBIT 12.4 Costco versus Peer Group: Working Capital in Days on HandNumber of days of revenues or cost of sales¹

	Costco			Peer Group		
	2016	2017	2018	2016	2017	2018
Operating cash	7.2	7.0	7.0	6.7	6.1	6.1
Accounts receivable, net	3.8	3.8	4.0	4.2	4.2	4.6
Inventory	31.7	30.7	30.9	54.3	52.3	52.7
Other current assets	0.8	0.8	0.8	1.5	1.4	1.6
Operating current assets ²	39.3	38.2	38.6	49.6	50.1	51.7
Accounts payable	29.5	28.1	30.9	48.2	50.6	54.4
Accrued salaries and benefits	7.8	7.5	7.3	4.8	5.3	5.7
Other current liabilities ³	12.3	13.2	13.8	12.8	12.1	11.8
Operating current liabilities ²	45.7	45.1	48.0	51.7	57.1	57.7
Working capital	(6.4)	(6.9)	(9.3)	(2.1)	(7.0)	(6.0)

¹ Days in inventory and accounts payable computed using cost of sales. Everything else computed using revenues. Measured using beginning- and end-of-year working capital account.

² Operating current assets and operating current liabilities do not equal the sum of individual accounts. Instead they are denoted in days of revenue.

³ Other current liabilities for Costco include accrued member rewards and deferred membership fees.

as long as they do at its peers'. Costco also has lower accounts payable days (30.9 versus 54.4 in 2018). This means it pays its suppliers faster, perhaps to get better prices.

Operating Analysis Using Nonfinancial Drivers In an external analysis, ratios are often confined to financial performance. If you are working from inside a company, however, or if the company releases operating data, link operating drivers directly to return on invested capital. By evaluating the operating drivers, you can better assess whether any differences in financial performance between competitors are sustainable.

Consider airlines, which are required for safety reasons to release a tremendous amount of operating data. Exhibit 12.5 details financial and operating data for two U.S. carriers we'll refer to as Airline A and Airline B between 2016 and 2018. Operating statistics include the number of employees, measured using full-time equivalents, and available seat-miles (ASMs), the common measurement of capacity for U.S. airlines.

Exhibit 12.6 transforms the data presented in Exhibit 12.5 into the operating-margin branch on the ROIC tree. Operating margin (operating profit divided by revenues) equals 9.1 percent for Airline A and 9.4 percent for Airline B. For airlines, operating margin is driven by three primary accounts: aircraft fuel, labor expenses, and other expenses. At first glance, it appears that Airlines A and B have similar labor costs. Labor expenses as a percentage of revenues average 27.7 percent for Airline A and 26.7 percent for Airline B. But

EXHIBIT 12.5 **Airline A and Airline B: Financial and Operating Statistics**

\$ million						
	Airline A			Airline B		
	2016	2017	2018	2016	2017	2018
Revenues	25,591	26,449	28,912	4,609	4,908	5,361
Aircraft fuel and related taxes	(4,069)	(4,839)	(6,515)	(752)	(954)	(1,329)
Salaries and related costs	(7,123)	(7,659)	(8,021)	(1,189)	(1,321)	(1,431)
Other operating expenses	(10,836)	(11,258)	(11,731)	(1,786)	(1,938)	(2,094)
Operating profit	3,562	2,693	2,645	882	695	506
Operating statistics						
Employees, full-time equivalent	64,400	62,860	61,600	11,190	12,197	12,787
Available seat-miles, millions	177,513	183,670	192,683	37,534	39,205	41,917

this statistic is misleading. To see why, disaggregate the ratio of labor expenses to revenues using available seat-miles (ASMs):

$$\frac{\text{Labor Expenses}}{\text{Revenues}} = \left(\frac{\text{Labor Expenses}}{\text{ASMs}} \right) \div \left(\frac{\text{Revenues}}{\text{ASMs}} \right)$$

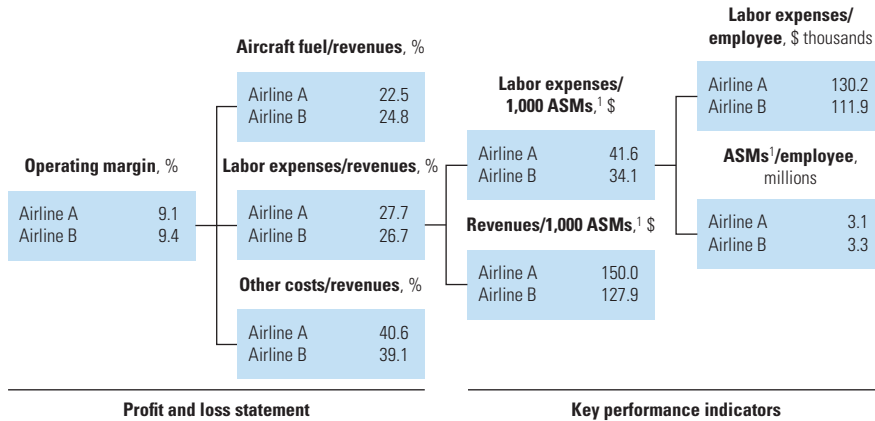
The ratio of labor expenses to revenues is a function of labor expenses per ASM and revenues per ASM. Labor expenses per ASM are the labor costs required to fly one ASM, and revenues per ASM represent the average price charged per ASM. Although Airlines A and B have similar ratios of labor expenses to revenues, they have different operating models. Airline B has an 18 percent advantage in labor cost per ASM (\$34.1 per thousand ASMs versus \$41.6 for Airline A). But what Airline A loses in labor costs, it recovers with higher prices. Because of its locations and reach, especially internationally, Airline A can charge an average price 17 percent higher than Airline B (\$150.0 per thousand ASMs versus \$127.9 per thousand ASMs).

But what is driving this differential in labor expenses per ASM? Are Airline B's employees more productive? Or are they paid less? To answer these questions, disaggregate labor expenses to ASMs, using the following equation:

$$\frac{\text{Labor Expenses}}{\text{ASM}} = \left(\frac{\text{Labor Expenses}}{\text{Employees}} \right) \div \left(\frac{\text{ASMs}}{\text{Employees}} \right)$$

Two elements drive labor expenses per ASM: the first term represents the average salary per full-time employee; the second measures the productivity of each full-time employee (millions of ASMs flown per employee). The boxes on the right side of Exhibit 12.6 report the calculations for this equation. The

EXHIBIT 12.6 Operating Drivers of Labor Expenses to Revenues, 2018



¹ Available seat-miles (ASMs) are the standard unit of measure for the U.S. airline industry.

average salary is 16.4 percent higher for Airline A, while productivity per mile is 4.6 percent lower. Although the salary differential appears significant, it is quite small in 2018 compared with the early 2000s, when average salaries differed by a factor of almost 2. Furthermore, the differences in productivity can be driven by different route structures and the level of service provided.

Analyzing performance using operating drivers gives additional insight into the competitive differences among airlines. But the analysis is far from done. In fact, a thoughtful analysis will often raise more questions than it answers. For instance, can the salary difference between Airline A and Airline B be explained by the mix of employees (pilots are more expensive than gate personnel) or the location of the employees (the East and West coasts are more expensive than the Midwest)? Each of these analyses will provide additional insight into each carrier type's ability to survive and prosper.

ANALYZING REVENUE GROWTH

Chapter 3 showed that ROIC, cost of capital, and growth in cash flows drive a company's value. By analyzing historical revenue growth, you can assess the potential for growth in the future.

The calculation of year-to-year revenue growth is straightforward, but the results can be misleading. Three prime culprits distort revenue growth: the effects of changes in currency values, mergers and acquisitions, and changes in

EXHIBIT 12.7 **Compass and Sodexo: Revenue Growth Analysis**

%	Compass			Sodexo		
	2016	2017	2018	2016	2017	2018
Persistent revenue	5.0	4.0	5.5	2.0	2.5	2.0
Rugby World Cup	—	—	—	0.5	(0.6)	—
Organic revenue growth	5.0	4.0	5.5	2.5	1.9	2.0
Currency effects	5.4	11.3	(4.6)	(0.4)	(0.8)	(5.9)
53-week year in United States	—	—	—	—	0.7	(0.4)
Acquisitions and divestitures	1.1	(0.2)	0.9	0.1	0.4	2.9
Reported revenue growth	11.5	15.1	1.8	2.2	2.2	(1.4)

accounting policies. Strip out any distortions created by these effects to arrive at a better forecast of organic revenue growth.

Exhibit 12.7 demonstrates how misleading raw year-to-year revenue growth figures can be. Compass (based in the United Kingdom) and Sodexo (based in France) are global providers of canteen services in businesses, health systems, schools, and sporting venues. As shown in the bottom line of the exhibit for 2017, total revenues at Compass grew by 15.1 percent, and revenues at Sodexo grew by just 2.2 percent. The difference in growth rates appears dramatic but is driven primarily by changes in currency values (pounds sterling versus euros), not by long-term stable organic revenue growth. When we strip out these and other distortions, we see that like-for-like organic revenue growth at Compass (4.0 percent) still exceeded Sodexo's revenue growth (1.9 percent), but by a much smaller amount.

In general, for large multinationals, swings in currency values and changes in corporate portfolios can make historical revenue growth extremely volatile, so benchmarking is difficult. At Compass, reported revenue growth fell from a high of 15.1 percent in 2017 to just 1.8 percent in 2018. This stands in stark contrast to the company's relatively stable organic revenue growth: between 4.0 and 5.5 percent over the same time period.

The next three sections discuss in detail each of the major sources of distortions— changes in currency values, mergers and acquisitions, and changes in accounting policies. For each, we consider its effect on performance measurement, forecasting, and, ultimately, valuation.

Currency Effects

Multinational companies conduct business in many currencies. At the end of each reporting period, these revenues are converted to the home currency of the reporting company. If foreign currencies are rising in value relative to the

company's home currency, this translation at better rates will lead to higher revenue numbers. Thus, a rise in revenue may not reflect increased pricing power or greater quantities sold, but simply depreciation in the company's home currency.

Compass and Sodexo are two companies exposed to foreign currency. The companies have similar geographic mixes, with nearly half of each company's revenues coming from North America. Since each company translates U.S. dollars into a different currency for its consolidated financial statements, however, exchange rates will affect each company quite differently.

Compass translates U.S. dollars from its North American business into pounds. Given the weakening of the pound against the U.S. dollar (\$1.51 per pound in 2015 versus \$1.30 per pound by 2017), Compass reported an increase in revenues of 5.4 percent in 2016 and 11.3 percent in 2017 attributable to the weakening pound, shown as "currency effects" in Exhibit 12.7. For Sodexo, exchange rates had the opposite effect. As the euro strengthened slightly against the dollar, Sodexo translated revenue from North America into fewer euros, leading to a 0.4 percent drop in euro-denominated revenues in 2016 and a 0.8 percent drop in 2017. Note how movements that helped Compass in 2016 and 2017 reversed themselves in 2018. Failing to acknowledge these currency movements can lead to a critical misunderstanding of a global company's ability to grow organically.

Mergers and Acquisitions

Growth through acquisition may have very different effects on value creation than internal growth does because of the sizable premiums a company must pay to acquire another company. Therefore, it is important to understand how companies have been generating historical revenue growth: through organic means or through acquisition.

Many large companies provide data tables such as the ones for Compass and Sodexo in Exhibit 12.7. Without voluntary disclosure, stripping the effect of acquisitions from reported revenues can be difficult. Unless an acquisition is deemed material by the company's accountants, company filings do not need to detail or even report the acquisition. For larger acquisitions, a company will report pro forma statements that recast historical financials as though the acquisition were completed at the beginning of the fiscal year. Organic revenue growth, then, should be calculated using the pro forma revenue numbers.³ If the target company publicly reports

³For example, Sodexo purchased Centerplate in November 2017. Since 2018 includes a full year of revenue from the Centerplate acquisition and 2017 does not, the company's consolidated revenue cannot be compared with the prior year's revenue without adjustment.

EXHIBIT 12.8 **Effect of Acquisitions on Revenue Growth**

\$ million

	Year				
	1	2	3	4	5
Revenue by company					
Acquiring company	100.0	110.0	121.0	133.1	146.4
Target company	20.0	22.0	24.2	26.6	29.3
Consolidated revenues					
Revenue of acquirer	100.0	110.0	121.0	133.1	146.4
Revenue from target			14.1	26.6	29.3
Consolidated revenues ¹	100.0	110.0	135.1	159.7	175.7
Growth rates of acquirer, %					
Reported growth ¹		10.0	22.8	18.2	10.0
Organic growth		10.0	10.0	10.0	10.0

¹ Only consolidated revenues are reported in a company's annual report.

its own financial data, you can construct pro forma statements manually by combining revenue of the acquirer and target for the prior year. But beware: the bidder will include partial-year revenues from the target for the period after the acquisition is completed. To remain consistent from year to year, reconstructed prior years also must include only partial-year revenue.

Exhibit 12.8 presents the hypothetical purchase of a target company in the seventh month of year 3. Both the parent company and the target are growing organically at 10 percent per year. Whereas the individual companies are growing organically at 10 percent, consolidated revenue growth is reported at 22.8 percent in year 3 and 18.2 percent in year 4.

To create an internally consistent comparison for years 3 and 4, adjust the prior year's consolidated revenues to match the current year's composition. To do this, add seven months of the target's year 2 revenue ($7/12 \times \$22$ million = \$12.8 million) to the parent's year 2 revenue (\$110.0 million). This leads to adjusted year 2 revenues of \$122.8 million, which matches the composition of year 3. To compute an organic growth rate, divide year 3 revenues (\$135.1 million) by adjusted year 2 revenues (\$122.8 million) to get the correct 10 percent organic growth of the two companies.

Even though the acquisition occurs in year 3, the revenue growth rate for year 4 also will be affected by the acquisition. Year 4 contains a full year of revenues from the target. Therefore, to estimate year 4 organic growth, you must increase year 3 revenue by five months of target revenue ($5/12 \times \$24.2$ million = \$10.1 million).

Accounting Changes and Irregularities

Each year, the Financial Accounting Standards Board (FASB) in the United States and the International Accounting Standards Board (IASB) make recommendations concerning the financial treatment of certain business transactions through either formal standards or topic notes issued by assigned task forces. Changes in a company's revenue recognition policy can significantly affect revenues during the year of adoption, distorting the one-year growth rate.⁴ You therefore need to eliminate their effects in order to understand real historical revenue trends.

Consider the new revenue recognition standards that replaced existing IFRS and GAAP revenue rules in 2017.⁵ These standards introduced a requirement that companies follow a five-step process to determine the allocation of revenue over the life of a contract, implied or written. In some cases, initiating this process caused revenues to be delayed to later in the contract, causing a one-time drop in like-for-like revenues. For example, automobile companies that provide free maintenance saw a one-time drop as revenues were delayed. Other industries, including cell phone providers, experienced a one-time increase in revenue as cell phone equipment sales can now be recognized immediately, rather than over the life of the contract.⁶

If an accounting change is material, a company will document the change in its section on management discussion and analysis (MD&A). For instance, as shown in Exhibit 12.7, Sodexo specifically called attention to an unusual 53-week year in 2017. The longer time period in 2017 artificially raised 2017 growth rates while lowering 2018 growth rates.

Decomposing Revenue Growth to Develop an Integrated Perspective of Growth Drivers

Once you have removed the effects of mergers and acquisitions, currency translations, and accounting changes from the year-to-year revenue growth numbers, analyze organic revenue growth from an operational perspective. The most standard breakdown is:

$$\text{Revenues} = \frac{\text{Revenues}}{\text{Units}} \times \text{Units}$$

⁴ Revenue recognition changes can also affect margins and capital turnover ratios. They will not, however, affect free cash flow.

⁵ ASC 606 and IFRS 15, "Revenue from Contracts with Customers," was issued jointly by the FASB and IASB on May 28, 2014. Implementation began in 2017.

⁶ F. Norris, "New Standards for Companies' Revenue Accounting Will Begin in 2017," *New York Times*, May 28, 2014.

Using this formula, determine whether prices or quantities are driving growth. Do not, however, confuse revenue per unit with price; they can be different. If revenue per unit is rising, the change could be due to rising prices, or the company could be shifting its product mix from low-price to high-price items.

The operating statistics that companies choose to report (if any) depend on the industry's norms and competitors' practices. For instance, most retailers provide information on the number of stores they operate, the number of square feet in those stores, and the number of transactions they conduct annually. By relating different operating statistics to total revenues, it is possible to build a deeper understanding of the business.

Consider this retailing standard:

$$\text{Revenues} = \frac{\text{Revenues}}{\text{Stores}} \times \text{Stores}$$

Exhibit 12.9 reports disguised operating statistics for two big-box retailers we'll call Delta and Gamma. Using the operating statistics reported in Exhibit 12.9, we discover that Delta has more stores than Gamma and generates more revenue per store (\$47 million per store for Delta in 2018 versus \$37 million for Gamma). Using the three operating statistics, it is possible to build ratios on revenues per store, transactions per store, square feet per store, dollars per transaction, and number of transactions per square foot.

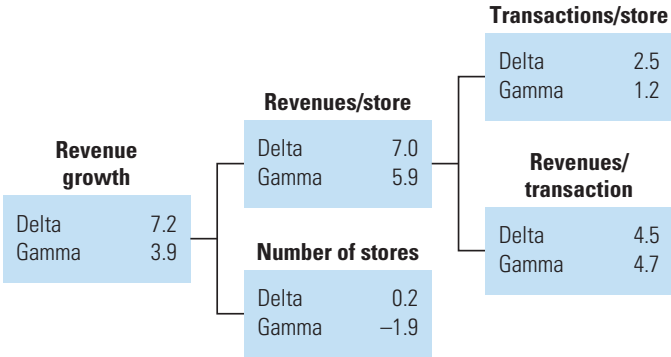
Although operating ratios are powerful in their own right, what can really change one's thinking about performance is how the ratios change over time. Exhibit 12.10 organizes each ratio based on Exhibit 12.9 into a tree. Rather than report a calculated ratio, such as revenues per store, however, we report the growth in the ratio over the period analyzed and relate this back to the growth in revenue.

EXHIBIT 12.9 Hypothetical Retailers: Operating Statistics

Reported	Delta			Gamma		
	2016	2017	2018	2016	2017	2018
Revenues, \$ million	51,081	54,488	58,430	35,109	37,054	38,507
Average number of stores	1,229	1,232	1,234	1,076	1,070	1,050
Number of transactions, millions	834	853	875	510	515	511
Average square footage, millions	90	90	90	85	86	85
Derived						
Revenues per store, \$ million	42	44	47	33	35	37
Transactions per store, thousands	678	692	709	474	481	487
Revenues per transaction, \$	61	64	67	69	72	75

EXHIBIT 12.10 **Hypothetical Retailers: Organic Revenue Growth Analysis, 2018**

Growth rates, %



As the exhibit demonstrates, Delta grew faster than Gamma, because Gamma closed stores while Delta slightly increased the number of stores, and Delta grew revenues per store faster than Gamma. Growth in revenues per store is the key driver for these two companies, because the category is near full penetration. This growth in same-store sales is extremely important, to the point that financial analysts have a special name for growth in revenue per store: *comps*, shorthand for comparables, or year-to-year same-store sales.⁷ Why is this revenue growth important? First, the number of stores to open is an investment choice, whereas same-store sales growth reflects each store's ability to compete effectively in its local market. Second, new stores require large capital investments, whereas growth in comps requires little incremental capital. Hence, same-store sales growth comes with higher capital turnover, higher ROIC, and greater value creation.

Moving farther right in the tree, we gain additional insight into what has been driving same-store sales for each company. Delta also generated more foot traffic, increasing transactions per store at 2.5 percent versus Gamma's 1.2 percent. Revenues per transaction grew at the same rate for the two stores.

CREDIT HEALTH AND CAPITAL STRUCTURE

To this point, we have focused on the operating performance of the company and its ability to create value. We have examined the primary drivers of

⁷ In Exhibit 12.10, we present the change in revenues per store. This value differs from comparable-store sales reported by each company, which includes only stores that were open for at least 13 months.

value: a company's return on invested capital and organic revenue growth. In the final step of historical analysis, we focus on how the company has financed its operations. What proportion of invested capital comes from creditors instead of from equity investors? Is this capital structure sustainable? Can the company survive an industry downturn? How much cash, if any, has been distributed to shareholders?

To assess a company's capital structure, conduct four analyses. First, examine liquidity using coverage ratios. Liquidity measures the company's ability to meet short-term obligations, such as interest expenses and rental payments. Next, evaluate leverage using debt to EBITDA and debt to value. Leverage measures the company's ability to meet obligations over the long term. To evaluate equity, measure the payout ratio and operating value to EBITDA. The payout ratio measures the percentage of income being sent to shareholders. Operating value to EBITDA measures shareholders' future expectations of financial performance.

This section introduces the tools for evaluating a company's capital structure. Chapter 33 examines how capital structure decisions must be an integral part of a company's operating strategy and its plan for how it will return cash to shareholders.

Measuring Liquidity Using Coverage Ratios

To estimate the company's ability to meet short-term obligations, analysts use ratios that incorporate three measures of earnings:

1. Earnings before interest, taxes, and amortization (EBITA)
2. Earnings before interest, taxes, depreciation, and amortization (EBITDA)
3. Earnings before interest, taxes, depreciation, amortization, and rental expense (EBITDAR)

With the first two earnings measures, you can calculate interest coverage. To do this, divide either EBITA or EBITDA by interest. The first coverage ratio, EBITA to interest, measures the company's ability to pay interest using profits without cutting capital expenditures intended to replace depreciating equipment. The second ratio, EBITDA to interest, measures the company's ability to meet short-term financial commitments using both current profits and the depreciation dollars earmarked for replacement capital. Although EBITDA provides a good measure of the short-term ability to meet interest payments, most companies cannot compete effectively without replacing worn assets.

An alternative is to divide EBITDAR by the sum of interest expense and rental expense. Like the interest coverage ratio, the EBITDAR ratio measures the company's ability to meet its known future obligations, including the effect of operating leases. For many companies, especially retailers and airlines,

EXHIBIT 12.11 **Costco versus Peer Group: Measuring Coverage**

\$ million

	Costco			Peer Group		
	2016	2017	2018	2016	2017	2018
EBITA	4,111	4,480	4,737			
EBITDA	5,481	5,917	6,229			
EBITDAR ¹	5,739	6,182	6,497			
Interest	134	159	150			
Rental expense	258	265	268			
Interest plus rental expense	392	424	418			
Coverage ratios						
EBITA/interest	30.7	28.2	31.6	9.0	8.3	15.3
EBITDA/interest	40.9	37.2	41.5	13.1	12.7	19.8
EBITDAR/interest plus rental expense	14.6	14.6	15.5	5.2	4.7	4.8
Debt multiples						
Debt to EBITA	1.71	1.56	1.53	2.21	2.35	2.40
Debt to EBITDA	1.28	1.18	1.16	1.53	1.53	1.67
Debt plus leases to EBITDAR	1.67	1.53	1.49	2.55	2.63	2.72

¹ Earnings before interest, taxes, depreciation, amortization, and rental expense.

including rental expenses is a critical part of understanding the financial health of the business.

Returning to our previous example of Costco and its peers, Exhibit 12.11 presents their financial data and coverage ratios. For 2018, Costco's coverage ratio of EBITA to interest equaled 31.6 times, whereas its peers had a ratio of 15.3 times. By most standards, Costco has very little debt, which is reflected in its extremely high AA– rating by Standard & Poor's. Costco's peers also have small amounts of leverage and high debt ratings, but to a lesser degree.

Measuring Leverage

Over the past decade, interest rates have dropped to unprecedented lows, making interest coverage ratios uncharacteristically high. To evaluate leverage in this low-interest-rate environment, many analysts are now measuring and evaluating debt multiples such as debt to EBITDA or debt to EBITA. Given its much larger denominator, debt to EBITDA tends to be more stable, making assessments over time much clearer. The ratio also does a better job of teasing out companies that are exposed to rollover risk and widening default spreads, neither of which is captured when interest rates are extremely low.

A second reason the debt-to-EBITDA measure has gained in popularity involves the increased use of convertible securities. Many convertibles compensate through the potential conversion to equity rather than interest, making

interest coverage ratios artificially high. By using the debt-to-EBITDA ratio, one can build a more comprehensive picture of the risk of leverage.

A variation of these debt multiples is the multiple of debt plus leases to EBITDAR. This multiple works best for companies with extensive operating leases, such as airlines and retailers.

To better understand the power—and danger—of leverage, consider the relationship between return on equity (ROE) and return on invested capital (ROIC):

$$\text{ROE} = \text{ROIC} + [\text{ROIC} - (1 - T)k_d] \frac{D}{E}$$

As the formula demonstrates, a company's ROE is a direct function of its ROIC, its spread of ROIC over its after-tax cost of debt (k_d), and its book-based debt-to-equity ratio (D/E). Consider a company that is earning an ROIC of 10 percent and has an after-tax cost of debt of 5 percent. To raise its ROE, the company can either increase its ROIC (through operating improvements) or increase its debt-to-equity ratio (by swapping debt for equity). Although each strategy can lead to an identical change in ROE, increasing the debt-to-equity ratio makes the company's ROE more sensitive to changes in operating performance (ROIC). Thus, while increasing the debt-to-equity ratio can increase ROE, it does so by increasing the risks faced by shareholders.

To assess leverage, measure the company's (market) debt-to-equity ratio over time and against peers. Does the leverage ratio compare favorably with the industry? How much risk is the company taking? Chapter 33 offers in-depth answers to these and other questions about the use of debt to finance operations.

Payout Ratio

The dividend payout ratio equals total common dividends divided by net income available to common shareholders. We can better understand the company's financial situation by analyzing the payout ratio in relation to its cash flow reinvestment ratio:

- If the company has a high dividend payout ratio and a reinvestment ratio greater than 1, then it must be borrowing money to fund negative free cash flow, to pay interest, or to pay dividends. But is this sustainable?
- A company with positive free cash flow and low dividend payout is probably paying down debt (or accumulating excess cash). In this situation, is the company passing up the valuable tax benefits of debt or hoarding cash unnecessarily?

Applying these questions to Costco, we find that from 2015 to 2019, Costco generated \$14.7 billion in NOPAT, paid \$1.1 billion in interest, and returned \$9.5 billion to shareholders in dividends.

Valuation Metrics

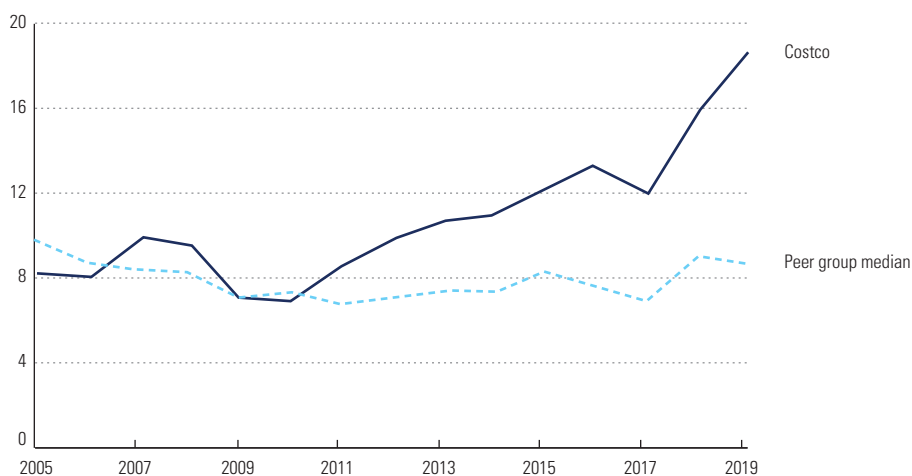
To conclude your assessment of capital structure, measure the shareholders' perception of future performance by calculating a market multiple. To build a market multiple, divide core operating value (defined in Chapter 10 as enterprise value less the market value of nonoperating assets, such as excess cash and nonconsolidated subsidiaries) by a normalizing factor, such as revenue, EBITA, or the book value of invested capital. By comparing the multiple of one company versus another, you can examine how the market perceives the company's future relative to other companies.

Exhibit 12.12 presents the operating-value-to-EBITDA multiples for Costco and its peers between 2005 and 2019. Although Costco traded in line with its peers from 2005 to 2011, its multiple has since increased substantially, while the multiple for its peers remained the same. We infer that Costco's higher multiple is driven by its stronger revenue growth and enduring higher ROIC.

While operating value to EBITDA is the most common measure of valuation, other measures, including operating value to EBITA and operating value to NOPAT, often provide helpful insights as well. For more on how to create and interpret valuation multiples, see Chapter 18.

EXHIBIT 12.12 **Costco versus Peer Group: Operating Value to EBITDA**

Multiple of EBITDA



Note: Operating value equals enterprise value less the book value of nonoperating assets.

GENERAL CONSIDERATIONS

Although it is impossible to provide a comprehensive checklist for analyzing a company's historical financial performance, here are some guidelines to keep in mind:

- Look back as far as possible (at least ten years). Long time horizons will allow you to determine whether the company and industry tend to revert to some normal level of performance and whether short-term trends are likely to be permanent.
- Disaggregate value drivers—both ROIC and revenue growth—as far as possible. If possible, link operational performance measures with each key value driver.
- If there are any radical changes in performance, identify the source. Determine whether the change is temporary, permanent, or merely an accounting effect.
- If possible, perform your analysis on a fine-grained level, not just on the company as a whole. Real insight comes from analysis of individual business units, product lines, and, if the data exist, even customers.

With historical analysis complete, we now have the appropriate context to build a robust set of forecasts, a critical ingredient of any valuation.

REVIEW QUESTIONS

1. Exhibit 12.13 presents historical data for ShipCo, a manufacturer of ships. Based on the data, how has ShipCo performed over the past five years? What is its ROIC for years 2–5 (using average invested capital)? What is its operating margin? What is its capital turnover? Is it creating more or less value over time? Assume an operating tax rate of 30 percent and a cost of capital of 9 percent.

EXHIBIT 12.13 **ShipCo: Income Statement and Balance Sheet**

\$ million

	Year 1	Year 2	Year 3	Year 4	Year 5		Year 1	Year 2	Year 3	Year 4	Year 5
Income statement						Balance sheet					
Revenues	530.0	556.5	601.0	661.1	674.3	Operating cash	10.6	11.1	12.0	13.2	13.5
Cost of sales	(344.5)	(364.5)	(396.7)	(439.6)	(451.8)	Excess cash and marketable securities	102.8	108.9	111.2	118.2	139.8
Selling, general, and administrative	(79.5)	(86.3)	(87.1)	(92.6)	(107.9)	Accounts receivable	79.5	80.7	93.2	99.2	97.8
Depreciation	(15.9)	(16.7)	(18.0)	(19.8)	(20.2)	Inventory	169.6	183.6	204.3	231.4	242.8
EBIT	90.1	89.0	99.2	109.1	94.4	Current assets	362.5	384.3	420.7	462.0	493.8
Interest expense	(7.5)	(7.5)	(7.5)	(7.5)	(7.5)	Property, plant, and equipment	206.7	219.8	240.4	267.8	276.5
Gain/(loss) on sale of assets	—	(10.0)	—	—	—	Equity investments	180.0	180.0	180.0	180.0	180.0
Earnings before taxes	82.6	71.5	91.7	101.6	86.9						
Taxes	(24.8)	(21.5)	(27.5)	(30.5)	(26.1)	Total assets	749.2	784.2	841.1	909.8	950.3
Net income	57.8	50.1	64.2	71.1	60.8						
Dividends	17.3	17.0	18.0	17.8	18.3	Accounts payable	116.6	119.6	126.2	135.5	134.9
						Short-term debt	45.0	45.0	45.0	45.0	45.0
						Accrued expenses	90.1	89.0	93.2	99.2	97.8
						Current liabilities	251.7	253.7	264.4	279.7	277.6
						Long-term debt	105.0	105.0	105.0	105.0	105.0
						Common stock	100.0	100.0	100.0	100.0	100.0
						Retained earnings	292.5	325.5	371.7	425.1	467.6
						Total liabilities and equity	749.2	784.2	841.1	909.8	950.3

2. Using the data presented in Question 1, decompose operating margin and capital turnover. What has been occurring at ShipCo over the five-year period? What might be driving these results?
3. Decompose the ratio of operating working capital to sales into operating-cash days, accounts-receivable days, inventory days, accounts-payable days, and accrued-expenses days. (For comparability, use sales in all calculations.) Compare these results with those in Exhibit 12.14 for DefenseCo, a large diversified defense contractor. Which firm is showing a better trend in its management of its working capital? What might explain the differences in these variables?

EXHIBIT 12.14 **DefenseCo ROIC Decomposition**

	Year 2	Year 3	Year 4	Year 5
Operating metrics				
ROIC, %	18.5	18.2	18.4	18.3
After-tax operating margin, %	7.1	6.9	7.0	6.9
Capital turns, times	2.62	2.65	2.63	2.64
Cost of sales/sales, %	73.0	73.1	72.8	72.9
SG&A/sales, %	14.4	14.6	14.7	14.7
Operating working capital/sales, %	3.1	2.9	3.1	3.1
PP&E/sales, %	35.1	34.9	34.9	34.8
Asset management, days				
Operating cash	7.1	7.1	7.1	7.1
Accounts receivable	42.1	41.8	42.7	42.5
Inventory	92.3	91.6	91.8	92.6
Operating current assets	141.5	140.5	141.6	142.2
Accounts payable	70.2	70.1	70.4	71.1
Accrued expenses	60.0	60.0	59.8	59.8
Operating current liabilities	130.2	130.1	130.2	130.9
Operating working capital	11.3	10.4	11.3	11.3

4. You are an equity analyst and have computed the following figures for two manufacturers of paper products. The first, PaperPro, has NOPAT of \$325 million, invested capital without goodwill of \$2,500 million, and goodwill of \$950 million. The second, ExpertPaper, has NOPAT of \$750 million, invested capital without goodwill of \$6,000 million, and no goodwill. If the cost of capital for both firms is 10 percent, are the firms creating value in this year? Which company has the best financial performance?
5. PaperPro had previously acquired PaperExpo, which independently generates \$800 million per year in revenues, with no material growth. The consolidated revenues for PaperPro (post-acquisition) are \$1.5 billion in year 1, \$1.8 billion in year 2 (the year of the acquisition), and \$2.5 billion in year 3. If PaperPro closed the acquisition of PaperExpo on October 1 of year 2, what is the “apples-to-apples” organic growth for PaperPro in year 2 and year 3? How does this differ from reported revenue growth? Assume that PaperExpo’s revenues are consolidated into PaperPro’s only after the acquisition close and that the fiscal year closes for both companies on December 31 of each year.
6. Which interest coverage ratio, EBITDA to interest or EBITA to interest, will lead to a higher number? When is the EBITDA interest coverage ratio more appropriate than the EBITA ratio? When is the EBITA interest coverage ratio more appropriate than the EBITDA ratio?